

dmall

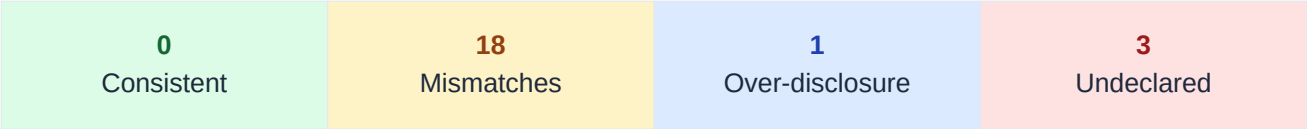
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|            |                 |
|------------|-----------------|
| Developer  | Donga developer |
| Genre      | SHOPPING        |
| Installs   | 50,000+         |
| Audit Date | 2026-04-26      |

Overall Risk Score

0.72

HIGH



Research prototype · Ongoing research | Policy: <https://dmall.co.kr/member/privacy.html>

## Executive Summary

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**dmail** declares 18 data type(s) collected and 0 shared in its Play Data Safety label, across 22 data type(s) analyzed. Our heterogeneous GNN audit model identifies **3 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **18 policy-vs-label mismatch(es)**. Overall risk score: **0.72 (HIGH)**.

### Top discrepancies by risk:

- **installed\_apps**: Undeclared Collection (confidence 0.99)
- **advertising\_id**: Undeclared Collection (confidence 0.97)
- **approximate\_location**: Undeclared Collection (confidence 0.86)

### Class definitions:

**CONSISTENT**: Label, policy, and SDK signals agree.

**POLICY\_LABEL\_MISMATCH**: Label discloses data not mentioned in policy.

**OVER\_DISCLOSURE**: Policy mentions data not declared in label.

**UNDECLARED\_COLLECTION**: SDK signals collection undisclosed in label and policy.

## Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

| Data Type                  | Class                 | Conf. | Evidence                            |
|----------------------------|-----------------------|-------|-------------------------------------|
| installed_apps             | Undeclared Collection | 0.99  | label:N policy:N sdk:Y (1, 2, 3 +5) |
| advertising_id             | Undeclared Collection | 0.97  | label:N policy:N sdk:Y (1, 2, 3 +5) |
| approximate_location       | Undeclared Collection | 0.86  | label:N policy:N sdk:Y (1, 2, 3 +5) |
| app_interactions           | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| device_id                  | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:Y (1, 2, 3 +5) |
| other_info                 | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| other_app_performance_data | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| email_address              | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:Y (1, 2, 3 +5) |
| in_app_search_history      | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| name                       | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:Y (1, 2, 3 +5) |
| phone_number               | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| user_ids                   | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| crash_logs                 | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:Y (1, 2, 3 +5) |
| diagnostics                | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:Y (1, 2, 3 +5) |
| photos                     | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| purchase_history           | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| address                    | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| financial_info             | Policy Label Mismatch | 1.00  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| contacts                   | Policy Label Mismatch | 0.99  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| user_payment_info          | Policy Label Mismatch | 0.99  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| files_and_docs             | Policy Label Mismatch | 0.99  | label:Y policy:N sdk:N (1, 2, 3 +5) |
| sms_mms                    | Over Disclosure       | 0.90  | label:N policy:Y sdk:N (1, 2, 3 +5) |

## Evidence Trails

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Detailed evidence for UNDECLARED\_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

### Discrepancy: installed\_apps — Undeclared Collection (confidence 0.99)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'installed\_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: advertising\_id — Undeclared Collection (confidence 0.97)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'advertising\_id' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: approximate\_location — Undeclared Collection (confidence 0.86)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'approximate\_location' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: app\_interactions — Policy Label Mismatch (confidence 1.00)

---

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'app\_interactions' collection, aligning it with the Play Data Safety label.

#### Discrepancy: device\_id — Policy Label Mismatch (confidence 1.00)

---

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'device\_id' collection, aligning it with the Play Data Safety label.

#### Discrepancy: other\_info — Policy Label Mismatch (confidence 1.00)

---

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'other\_info' collection, aligning it with the Play Data Safety label.

#### Discrepancy: other\_app\_performance\_data — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'other\_app\_performance\_data' collection, aligning it with the Play Data Safety label.

#### Discrepancy: email\_address — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'email\_address' collection, aligning it with the Play Data Safety label.

#### Discrepancy: in\_app\_search\_history — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'in\_app\_search\_history' collection, aligning it with the Play Data Safety label.

#### Discrepancy: name — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'name' collection, aligning it with the Play Data Safety label.

#### Discrepancy: phone\_number — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'phone\_number' collection, aligning it with the Play Data Safety label.

#### Discrepancy: user\_ids — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'user\_ids' collection, aligning it with the Play Data Safety label.

#### Discrepancy: crash\_logs — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'crash\_logs' collection, aligning it with the Play Data Safety label.

#### Discrepancy: diagnostics — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'diagnostics' collection, aligning it with the Play Data Safety label.

#### Discrepancy: photos — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'photos' collection, aligning it with the Play Data Safety label.

#### Discrepancy: purchase\_history — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'purchase\_history' collection, aligning it with the Play Data Safety label.

#### Discrepancy: address — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'address' collection, aligning it with the Play Data Safety label.

#### Discrepancy: financial\_info — Policy Label Mismatch (confidence 1.00)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'financial\_info' collection, aligning it with the Play Data Safety label.

#### Discrepancy: contacts — Policy Label Mismatch (confidence 0.99)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'contacts' collection, aligning it with the Play Data Safety label.

#### Discrepancy: user\_payment\_info — Policy Label Mismatch (confidence 0.99)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'user\_payment\_info' collection, aligning it with the Play Data Safety label.

#### Discrepancy: files\_and\_docs — Policy Label Mismatch (confidence 0.99)

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Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'files\_and\_docs' collection, aligning it with the Play Data Safety label.

#### Discrepancy: sms\_mms — Over Disclosure (confidence 0.90)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'sms\_mms' is genuinely collected. If yes, add to label; if no, remove from policy.

## Methodology

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PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED\_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: [https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5\\_model\\_card.md](https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md).

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

## Disclaimer

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This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.